Name: _____

WORKSHEET

# Vector properties

**1** Use a diagram to show that addition of the following vectors is commutative $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$.

**a** $(1, 4)$ and $(3, -2)$

**b** $(-2, -4)$ and $(0, 3)$

**c** $(-5, 5)$ and $(2, -6)$

**d** $(6, -4)$ and $(-5, 7)$

**e** $(4, -6)$ and $(-1, 3)$

**f** $(9, -4)$ and $(-5, -2)$

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**g** $(7, 7)$ and $(-3, -8)$

**h** $(3, -5)$ and $(-6, 2)$

**2** Find the additive inverse of the following vectors

**a** $(6, 2)$

**b** $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$

**c** $(7, 90°)$

**d** $-3\mathbf{i} + \mathbf{j}$

**e** $(8, -11)$

**f** $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$

**g** $(3.96, 193°)$

**h** $12\mathbf{i} - 9\mathbf{j}$

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**3** Show that vector addition is associative using the vectors below $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$.

**a** $\mathbf{a} = (6, -1)$ $\mathbf{b} = (-3, 9)$ $\mathbf{c} = (0, 4)$

**b** $\mathbf{a} = (2, 90°)$ $\mathbf{b} = (8, 180°)$ $\mathbf{c} = (5, 270)$

**c** $\mathbf{a} = \mathbf{i} - 5\mathbf{j}$ $\mathbf{b} = 4\mathbf{i} + 7\mathbf{j}$ $\mathbf{c} = -3\mathbf{i} + 11\mathbf{j}$

**d** $\mathbf{a} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$ $\mathbf{c} = \begin{pmatrix} -4 \\ -8 \end{pmatrix}$

**4** Given $\mathbf{r} = 3$, $\mathbf{s} = -7$, $\mathbf{u} = (2, -1)$ and $\mathbf{v} = (-6, 4)$ show that:

**a** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

**b** $\mathbf{r}(\mathbf{u} + \mathbf{v}) = \mathbf{ru} + \mathbf{rv}$

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**c** $(\mathbf{r} + \mathbf{s})\mathbf{v} = \mathbf{rv} + \mathbf{sv}$

**d** $(\mathbf{rs})\mathbf{u} = \mathbf{r}(\mathbf{su})$

**e** $\mathbf{ru} + \mathbf{sv} = \mathbf{sv} + \mathbf{ru}$

**f** $\mathbf{ru} = \mathbf{cu} \Leftrightarrow \mathbf{r} = \mathbf{c}$

**5** Given $\mathbf{r} = -2$, $\mathbf{s} = 5$, $\mathbf{u} = (4, 250°)$ and $\mathbf{v} = (7, 125°)$ show that:

**a** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

**b** $\mathbf{r}(\mathbf{u} + \mathbf{v}) = \mathbf{ru} + \mathbf{rv}$

**c** $(\mathbf{r} + \mathbf{s})\mathbf{v} = \mathbf{rv} + \mathbf{sv}$

**d** $(\mathbf{rs})\mathbf{u} = \mathbf{r}(\mathbf{su})$

**e** $\mathbf{ru} + \mathbf{sv} = \mathbf{sv} + \mathbf{ru}$

**f** $\mathbf{ru} = \mathbf{cu} \Leftrightarrow \mathbf{r} = \mathbf{c}$

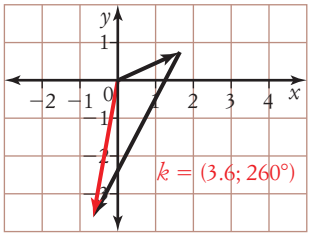
**6** Prove that $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ for vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

**7** Given $\mathbf{r}$ and $\mathbf{s}$ are scalars and $\mathbf{v}$ is a vector, prove that $(\mathbf{r} + \mathbf{s})\mathbf{v} = \mathbf{rv} + \mathbf{sv}$.

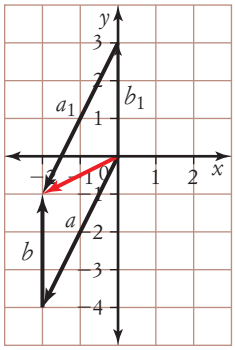
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## Answers

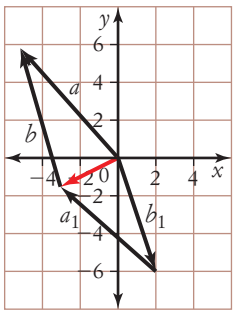
**1 a**



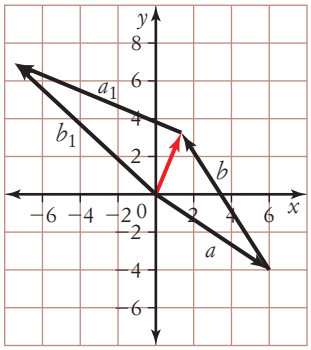

**b**



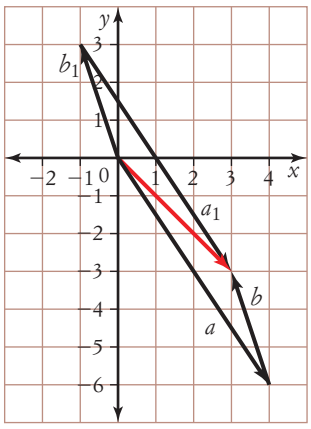

**c**



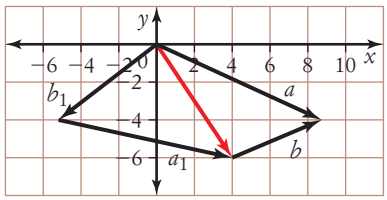

**d**



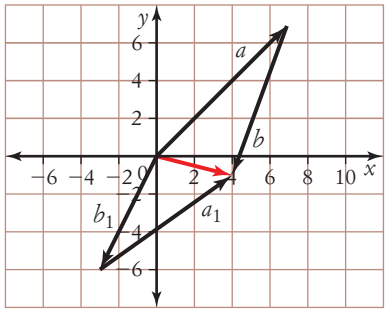

**e**



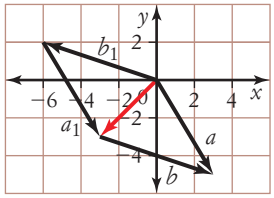

**f**




**g**




**h**




**2 a** $(-6, -2)$

**b** $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$

**c** $(7, 270°)$

**d** $3\mathbf{i} - \mathbf{j}$

**e** $(-8, 11)$

**f** $\begin{pmatrix} 0 \\ -7 \end{pmatrix}$

**g** $(3.96, 13°)$

**h** $-12\mathbf{i} + 9\mathbf{j}$

**3–5** Proofs